

Sentiment Analysis Extension

Problem Statement

Understanding customer opinions and emotional tone from text data is crucial for decision-making in domains like customer feedback, reviews, and social media.

Approach

- Built a **sentiment analysis pipeline** using:
 - NLP preprocessing (tokenization, cleaning, normalization)
 - Pre-trained language models (via Hugging Face where applicable)
- Supports classification into:
 - Positive
 - Negative
 - Neutral sentiments

Methodology

- Text data is cleaned and vectorized
- Models are trained or fine-tuned on labeled sentiment datasets
- Predictions include:
 - Sentiment label
 - Confidence score

Applications

- Customer feedback analysis
- Product review monitoring
- Social media sentiment tracking

Deployment

- Entire system deployed through **AI-Hub using RapidMiner**
- Images and data stored in **AWS S3**
- Inference handled via automated workflows
- Supports:
 - Real-time predictions
 - Error handling
 - Scalable deployment

Conclusion

This project delivers a **complete AI ecosystem** for:

- Image Classification
- Object Detection
- Sentiment Analysis

The key strength lies in its **flexibility, modularity, and deployability**, allowing users to:

- Choose models based on requirements
- Fine-tune on custom datasets
- Deploy seamlessly in production environments

It bridges the gap between **research-level models and real-world industrial**

applications.